

Notes

1. Introduction

- 1 U.S. Senate, *Trade Agreements Compliance Act*, Hearing before the Subcommittee on International Trade of the Committee on Finance, 101st Congress, 2nd Session, July 13, 1990, p. 30. Testimony of Andrew A. Procassini.
- 2 U.S. House of Representatives, *High Definition Television*, Hearing before the Subcommittee on Research and Development and the Subcommittee on Investigations of the Committee on Armed Services, 101st Congress, 1st Session, May 10, 1989, p. 66. Testimony of Pat Hubbard.
- 3 U.S. Senate, *Department of Defense Authorization for Appropriations for Fiscal Years 1990 and 1991*, Hearings before the Committee on Armed Services, 101st Congress, 1st Session, March 17; May 11, 16, 17, 31; June 2, 1989, p. 10. Testimony of Mel Levine (D-California).
- 4 The strategic-trade literature generally takes “high technology” to mean R&D intensity, which serves as an entry barrier and thus conveys information about the degree to which an industry is imperfectly competitive. Other characteristics further distinguish high-technology industries, as is discussed more fully in Chapter 2.
- 5 This logic obtains where this supplier also competes under conditions of increasing returns to scale.
- 6 James Brander and Barbara J. Spencer, “Export Subsidies and International Market Share Rivalry,” *Journal of International Economics* 18 (1985): 83–100.
- 7 This is to say that the United States and Europe have shied away from subsidizing a market presence in the HDTV and semiconductor industries, respectively, not that they have invested zero resources. See Chapters 5 and 4, respectively.
- 8 See, for example, Gene M. Grossman and Elhanan Helpman, “Trade Wars and Trade Talks,” *Journal of Political Economy* 103.4 (1995): 675–708; and John A. C. Conybeare, *Trade Wars: The Theory and Practice of International Commercial Rivalry* (New York: Columbia University Press, 1987).
- 9 Harry G. Johnson, *International Trade and Economic Growth: Studies in Pure Theory* (Cambridge, MA: Harvard University Press, 1967). See also Joanne Gowa and Edward D. Mansfield, “Power Politics and International

- Trade,” *American Political Science Review* 87.2 (1993): 408–20; and Conybeare, *Trade Wars*.
- 10 Joanne Gowa, “Rational Hegemons, Excludable Goods, and Small Groups: An Epitaph for Hegemonic Stability Theory?” *World Politics* 41.3 (1989): 307–24; and David A. Lake, *Power, Protection and Free Trade: International Sources of U.S. Commercial Strategy, 1887–1939* (Ithaca, NY: Cornell University Press, 1988).
 - 11 David A. Lake, “Leadership, Hegemony, and the International Economy: Naked Emperor or Tattered Monarch with Potential,” *International Studies Quarterly* 37.4 (1993): 473.
 - 12 Market power in the optimal tariff context implies that a state is a sufficiently large purchaser that it faces a less than perfectly elastic foreign supply curve. This means that the price of imports does not increase by the full amount of the tariff, part of the cost being passed on to foreign trade rivals. Market structure in the strategic-trade context raises questions about barriers to entry and exit and sources of increasing returns to scale. This is discussed more fully in Chapter 2.
 - 13 Paul R. Krugman, “Introduction,” in Krugman and Alasdair Smith (eds.), *Empirical Studies of Strategic Trade Policy* (Chicago: University of Chicago Press, 1994), p. 1–9; J. David Richardson, *Sizing-Up U.S. Export Disincentives* (Washington, DC: Institute for International Economics, 1993); Laura D’Andrea Tyson, *Who’s Bashing Whom? Trade Conflict in High-technology industries* (Washington, DC: Institute for International Economics, 1992); Cynthia A. Beltz, *High-Tech Maneuvers: Industrial Policy Lessons of HDTV* (Washington, DC: American Enterprise Institute, 1991); Michael E. Porter, *The Competitive Advantage of Nations* (New York: Free Press, 1990); Robert Wade, *Governing the Market: Economic Theory and the Role of Government in East Asian Industrialization* (Princeton: Princeton University Press, 1990); Jeffrey A. Hart, “Strategic Impacts of High Definition Television for U.S. Manufacturing” (Ann Arbor, MI: National Center for Manufacturing Sciences, 1989); Jeffrey A. Hart and Laura D’Andrea Tyson, “Responding to the Challenge of HDTV,” *California Management Review* 31.4 (1989): 132–45; Daniel I. Okimoto, *Between MITI and the Market: Japanese Industrial Policy for High Technology* (Stanford: Stanford University Press, 1989); and Hugh Patrick, “Japanese High Technology Policy in Comparative Context,” Columbia University Graduate School of Business, Center on Japanese Economy and Business Working Paper No. 1 (New York, 1986).
 - 14 Robert Ford and Win Swyker, “Industrial Subsidies in the OECD Economies,” *OECD Economic Studies* 15 (1990): 37–81; and Gene M. Grossman, “Promoting New Industrial Activities: A Survey of Recent Arguments and Evidence,” *OECD Economic Studies* 11 (1988): 87–125.
 - 15 See, for example, Beltz, *High-Tech Maneuvers*.
 - 16 Avinash Dixit, “How Should the United States Respond to Other Countries’ Trade Policies,” in Robert M. Stern (ed.), *U.S. Trade Policies in a Changing World Economy* (Cambridge, MA: MIT Press, 1987), p. 246.

- 17 See, for example, John Mark Hansen, *Gaining Access: Congress and the Farm Lobby, 1919–1981* (Chicago: University of Chicago Press, 1991); Cheryl Schoenhardt-Bailey, “Lessons in Lobbying for Free Trade in 19th Century Britain: To Concentrate or Not,” *American Political Science Review* 85.1 (1991): 37–58; and Vinod K. Aggarwal, Robert O. Keohane, and David B. Yoffie, “The Dynamics of Negotiated Protectionism,” *American Political Science Review* 81.2 (1987): 345–66.
- 18 See, for example, Gene M. Grossman and Elhanan Helpman, “Protection for Sale,” *American Economic Review* 84.4 (1994): 833–50; Arye L. Hillman, “Protection, Politics, and Market Structure,” in Elhanan Helpman and Assaf Razin (eds.), *International Trade and Trade Policy* (Cambridge, MA: MIT Press, 1991), pp. 118–40; Stephen P. Magee, William A. Brock, and Leslie Young, *Black Hole Tariffs and Endogenous Policy Theory: Political Economy in General Equilibrium* (Cambridge: Cambridge University Press, 1989); and Marc L. Busch and Eric Reinhardt, “Industrial Location and Protection: The Political and Economic Geography of U.S. Nontariff Barriers,” *American Journal of Political Science* (forthcoming).
- 19 Europe, in this case, being a shorthand for the four Airbus states, including Britain, France, Germany, and Spain.

2. The Argument

- 1 Paul R. Krugman, “Introduction,” in Krugman and Alastair Smith (eds.), *Empirical Studies of Strategic Trade Policy* (Chicago: University of Chicago Press, 1994), pp. 1–9.
- 2 For a review of the literature, see Klaus Stegemann, “Policy Rivalry among Industrial States: What Can We Learn from Models of Strategic Trade Policy?” *International Organization* 43.1 (1989): 73–100.
- 3 James A. Brander and Barbara J. Spencer, “Export Subsidies and International Market Share Rivalry,” *Journal of International Economics* 18 (1985): 83–100.
- 4 Gottfried Haberler, “Strategic Trade Policy and the New International Economics: A Critical Analysis,” in Ronald W. Jones and Anne D. Krueger (eds.), *The Political Economy of International Trade: Essays in Honor of Robert E. Baldwin* (Cambridge, MA: Blackwell, 1990), pp. 25–30; J. David Richardson, “The Political Economy of Strategic Trade Policy,” *International Organization* 44.1 (1990): 107–35; Jagdish Bhagwati, *Protectionism* (Cambridge, MA: MIT Press, 1988); and Gene M. Grossman and J. David Richardson, “Strategic Trade Policy: A Survey of Issues and Early Analysis,” Princeton University Special Papers in International Economics No. 15 (Princeton, 1985).
- 5 James A. Brander, “Rationales for Strategic Trade and Industrial Policy,” in Paul R. Krugman (ed.), *Strategic Trade Policy and the New International Economics* (Cambridge, MA: MIT Press, 1986), p. 44. Emphasis added.
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- 7 The social rate of return is the private rate of return plus the interindustry marginal cost reductions owing to these technology spillovers. See, for example, Jeffrey I. Bernstein and M. Ishaq Nadiri, "Interindustry R&D Spillovers, Rates of Return, and Production in High-Tech Industries," *American Economic Review* 78.2 (1988): 429.
 - 8 See Chapter 5.
 - 9 In the case studies, the state, as a rational unitary actor, is operationalized as the executive. This accords with studies that link the rational unitary state assumption to executive politics more generally, as where, for example, the executive monopolizes information or holds the same preferences as the legislature. The book speaks of states as rational unitary actors, rather than invoking the language of executive politics, because of its focus on international, not domestic politics, the latter being the focus of the competing explanation. For a discussion of rational unitary states and executive politics, see Helen V. Milner, *Interests, Institutions, and Information: Domestic Politics and International Relations* (Princeton: Princeton University Press, 1997), pp. 11–12. For a more general discussion of the utility maximization assumption, see Jeffrey A. Frieden, *Debt, Development, and Democracy: Modern Political Economy and Latin America, 1965–1985* (Princeton: Princeton University Press, 1991), pp. 17–19.
 - 10 Jean Tirole, "The Internal Organization of Government," *Oxford Economic Papers* 46 (1994): 1–29.
 - 11 Wesley M. Cohen and Richard C. Levin, "Empirical Studies of Innovation and Market Share," in Richard Schmalensee and Robert D. Willig (eds.), *Handbook of Industrial Organization*, Vol. 2 (Amsterdam: North Holland, 1989), pp. 1059–1107.
 - 12 Geoffrey Carliner, "Industrial Policies for Emerging Industries," in Krugman, *Strategic Trade Policy and the New International Economics*, p. 156.
 - 13 Lewis M. Branscomb, "The National Technology Policy Debate," in Branscomb (ed.), *Empowering Technology: Implementing a U.S. Strategy* (Cambridge, MA: MIT Press, 1993), pp. 1–35; Organisation for Economic Co-operation and Development, *Technology and the Economy: The Key Relationship* (Paris: OECD, 1992); and Eric von Hippel, *The Sources of Innovation* (New York: Oxford University Press, 1988).
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- Technology Competition* (Cambridge, MA: Harvard University Press, 1992); Richard J. Caballero and Richard K. Lyons, "External Effects and Europe's Integration," in L. Alan Winters and Anthony Venables (eds.), *European Integration: Trade and Industry* (Cambridge: Cambridge University Press, 1991), pp. 34–51; Cohen and Levin, "Empirical Studies of Innovation and Market Share"; Jeffrey I. Bernstein, "Costs of Production, Intra- and Interindustry R&D Spillovers: Canadian Evidence," *Canadian Journal of Economics* 21.2 (1988): 324–47; Bernstein and Nadiri, "Interindustry R&D Spillovers, Rates of Return, and Production in High-Tech Industries"; and Martin B. Zimmerman, "Learning Effects and the Commercialization of New Energy Technologies: The Case of Nuclear Power," *Bell Journal of Economics* 13.2 (1982): 297–310.
- 15 Organisation for Economic Co-operation and Development, *Technology and Industrial Performance* (Paris: OECD, 1996), p. 19.
 - 16 Annelee Saxenian, *Regional Advantage: Culture and Competition in Silicon Valley and Route 128* (Cambridge, MA: Harvard University Press, 1994).
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 - 21 Avinash Dixit, "How Should the United States Respond to Other Countries' Trade Policies?" in Stern, *U.S. Trade Policies in a Changing World Economy*, pp. 245–82.

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- 28 See Robert O. Keohane, *After Hegemony: Cooperation and Discord in the World Political Economy* (Princeton: Princeton University Press, 1984).
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- 31 Laura D'Andrea Tyson, *Who's Bashing Whom? Trade Conflict in High-Technology Industries* (Washington, DC: Institute for International Economics, 1992).
- 32 See Organisation for Economic Co-operation and Development, *Technology and Industrial Performance*, p. 37.
- 33 See, for example, Wendy L. Hansen and Kee Ok Park, "Nation-State and Pluralistic Decision Making in Trade Policy: The Case of the International Trade Administration," *International Studies Quarterly* 39.2 (1995): 181–212; Edward D. Mansfield and Marc L. Busch, "The Political Economy of Nontariff Barriers: A Cross-National Analysis," *International Organization* 49.4 (1995): 723–49; Daniel Treffer, "Trade Liberalization and the Theory of Endogenous Protection: An Econometric Study of U.S. Import Policy," *Journal of Political Economy* 101.1 (1993): 138–60; John A. C. Conybeare, "Voting for Protection: An Electoral Model of Tariff Policy," *International Organization* 45.1 (1991): 57–82; Stephen P. Magee, William A. Brock, and Leslie Young, *Black Hole Tariffs and Endogenous Policy Theory: Political Economy in General Equilibrium* (Cambridge: Cambridge University

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3. The Civil Aircraft Rivalry

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- 2 Office of Technology Assessment, "Competing Economies: America, Europe, and the Pacific Rim," OTA-ITE-498 (Washington, DC: GPO, 1991), p. 342.
- 3 Martin B. Lieberman, "The Learning Curve and Pricing in the Chemical Processing Industries," *Rand Journal of Economics* 15.2 (1984): 214.
- 4 Gernot Klepper, "Industrial Policy in the Transport Aircraft Industry," in Paul R. Krugman and Alasdair Smith (eds.), *Empirical Studies of Strategic Trade Policy* (Chicago: University of Chicago Press, 1994), p. 104. The 80% figure was provided by Airbus North America in a 1996 interview with Jonathan Schofield, chairman and CEO.
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- 6 "Deutsche Airbus to Begin Joining Wing of First A321 This Week," *Aviation Week and Space Technology*, July 13, 1992, p. 23.
- 7 David B. Yoffie, "Collision Course in Commercial Aircraft: Boeing-Airbus-McDonnell Douglas 1991 (A)," Harvard Business School Case No. 9-391-109 (Cambridge, MA, 1991), p. 3; and U.S. Department of Commerce, A

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 - 12 Oral Statement of Lawrence W. Clarkson, Corporate Vice President for Planning and International Development, The Boeing Company, before the House Committee on Public Works and Transportation, Subcommittee on Aviation, February 24, 1993, p. 3. Testimony courtesy of the Boeing Company.
 - 13 Interview with Schofield and Alan Boyd, former CEO of Airbus North America.
 - 14 Baldwin and Krugman, “Industrial Policy and International Competition in Wide-Bodied Jet Aircraft.”
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 - 16 Commission of the European Communities, “Demand Prospects for Civil Transport Aircraft” (Brussels: Commission of the European Communities, 1975), p. 15.
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 - 19 Derivative airplanes extend the range (i.e., Boeing’s 777-200ER) or seating capacity (i.e., Airbus’s A340-300) of a base model.
 - 20 See, for example, “Aerospatiale Wary of Superjumbo Project,” *Financial Times*, March 3, 1994, p. 4.
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 - 23 Testimony of Lawrence W. Clarkson, Corporate Vice President for Planning and International Development, The Boeing Company, before the Senate Finance Committee, June 2, 1992, p. 2. Testimony courtesy of the Boeing Company.
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- 49 U.S. Department of Commerce, 1994 *U.S. Industrial Outlook*, p. 27-13.

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- 52 "Aerospatiale Strives to Keep Stability in CAD/CAM Hardware, Software," *Aviation Week and Space Technology*, June 22, 1992, pp. 55–6.
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5. The High-Definition Television Rivalry

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6. Robotics, Superconductors, and Wheat

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7. Conclusion

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